# ELECTROTECHNICS & EMBEDDED SYSTEMS ENGINEER | KiCad PCB DESIGNER

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## PROFESSIONAL SUMMARY

Electrotechnics & Embedded Systems Engineer and Founder of Iotrixa (Arrˆ“tˆ' 1275 startup label recognized by the) show lm 714 moveto (Ministry of Hi

PCB routing, signal isolation, power electronics topologies, and real-time firmware development in Embedded C/C++ on STM32 ARM Cortex-M and ESP32 microcontrollers with FreeRTOS multi-threading.

## CORE ENGINEERING COMPETENCIES

**Hardware & PCB Design:** KiCad Schematic & Multi-Layer Layout, Ground Plane Isolation, EMI Reduction, AC-DC Buck Topologies

**Embedded Firmware:** STM32 (HAL / LL), ESP32 (C3 / S3 / WROOM), Embedded C/C++, FreeRTOS, PID Motor Control

**IoT, Telemetry & Protocols:** MQTT over TLS, RS-485 / Modbus RTU, I2C, SPI, UART, Wi-Fi Telemetry & Cloud Dashboards

**Prototyping & Bring-Up:** SMD Soldering, Digital Oscilloscope Signal Inspection, Logic Analyzers, Hardware Bench Testing

## ENGINEERING & STARTUP EXPERIENCE

### Founder & Lead Embedded Engineer — 2024 - Present

*Iotrixa -- Arrˆ“tˆ' 1275 Labelized Startup Project | Algiers, Algeria*

- Architected custom IoT hardware telemetry boards integrating ESP32-C3 RISC-V, AC-DC power, and opto-relays.
- Designed 1-layer & multi-layer PCBs in KiCad with dedicated ground planes, snubber networks, and transient protection.
- Implemented robust MQTT over TLS telemetry reporting real-time sensor data and remote actuation commands.
- Awarded official Arrˆ“tˆ' 1275 Startup Innovation Label by the Ministry of Higher Education & Scientific Research.

### Hardware & Embedded Control Engineer — 2024 - 2025

*National Robotics Tournament Team | Competitive Robotics Platform*

- Designed closed-loop PID velocity control firmware and high-current H-bridge MOSFET driver circuits.
- Integrated ultrasonic sensor matrices and quadrature optical encoders for real-time autonomous obstacle avoidance.
- Conducted extensive board bring-up, PWM tuning, and battery power management with 4S LiPo topologies.

## FEATURED HARDWARE & EMBEDDED PROJECTS

**Iotrixa Smart Telemetry & Relay Board** — *KiCad, ESP32-C3, MQTT, AC-DC Buck 230V to 5V*

- Engineered commercial IoT hardware board handling 10A 250VAC switching with isolated logic power domains.

**SafeBelt - STM32 Motor Controller & Safety Board** — *KiCad Multi-Layer, STM32F4, RS-485, Optocouplers*

- Sub-microsecond hardware overcurrent comparator shutoff circuit shielding STM32 from switching transients.

## EDUCATION & RECOGNITIONS

### Master / MCIL in Electrotechnics & Embedded Systems — Specialized Degree

*Faculty of Technology | Algeria*

- Curriculum: Industrial electrotechnics, power distribution, closed-loop motor control, and embedded systems.

### Arrˆ“tˆ' 1275 Startup Innovation Label (Iotrixa) — 2024 - 2025

*Ministry of Higher Education & Scientific Research*

- National startup designation recognizing Iotrixa as an innovative technology and IoT telemetry enterprise.